

iSpy-FRC - 7-Day Pre-Release Plan

Focused execution plan based on the provided Pre-Release Audit

Priority rule: Release blockers come first. Do not spend Day 7 polishing documentation while a licensing, data-loss,

platform-install, testing, or boot-resilience blocker remains.

Day 1 - Licensing + Platform Install

BLOCKER

~~Make the licensing decision first: either ship this release under AGPL-3.0, or finish enough isolation that [decision documented: code stays GPL-3.0; default .pt models keep Ultralytics AGPL-3.0]~~

~~the affected YOLO paths do not run Ultralytics in-process.~~

~~If keeping the current architecture, do not ship PolyForm Noncommercial.~~

~~Fix setup_systemd() so it does not hardcode user='pi'; detect/configure the service user.~~

Fresh-install and service-start test on a real Raspberry Pi and a real Orange Pi image. (deferred: needs hardware)

EXIT

CRITERIA

You have a documented license decision, service installation works on both target boards, and no known licensing blocker remains.

Day 2 - Config Durability + Boot Resilience

BLOCKER

~~Make iSpyConfig.save() atomic: write to a temporary file, then os.replace() the live config.~~

~~Apply the atomic-save fix to every config.save() call site.~~

~~Wrap each camera construction in _build_cameras_from_config with per-camera exception handling; log and skip bad cameras.~~

~~Implement a legacy-config migration path so upgrades do not require boot -f and data loss.~~

EXIT

CRITERIA

A bad camera cannot take down all cameras, power loss during a config write does not corrupt config.json, and an old config upgrades successfully.

Day 3 - Security Pass

BLOCKER

~~Add MAX_CONTENT_LENGTH to both Flask apps.~~

~~Check available disk space before large model/dataset uploads.~~

BLOCKER

~~Protect service_daemon.py start/stop/restart endpoints with an operator token or basic authentication.~~

~~Re-enable TLS verification in dataset.py.~~

~~Remove, disable, or clearly isolate the search-engine scraping fallback.~~

~~Choose a clear plugin-upload trust model: authenticated trusted-user code execution, or redesign the validation. Do not rely on the bypassable source blocklist as a security boundary.~~

EXIT

CRITERIA

Untrusted network users cannot remotely control the service, oversized uploads are rejected, TLS is verified, and the plugin trust model is explicit.

Day 4 - Concurrency + Resource Leaks

Fix the RollBack worker-thread leak when stop/start occurs and the sentinel cannot be queued.

Bound or document Camera._open_capture_bounded hung-thread behavior; add a maximum reconnect-thread cap if practical.

Make NetworkTables connection non-blocking/backgrounded so boot does not stall for up to 15 seconds.

Add a lock around iSpyConfig writes shared by background optimization threads and web handlers.

EXIT

CRITERIA

Repeated camera reconnects and recorder restarts do not continuously accumulate workers, config writes are serialized, and NetworkTables cannot block normal boot.

Day 5 - Automated + Manual Testing

BLOCKER

~~Restore a real unit-test subset; the current unit_tests.py is commented out and provides no regression protection.~~

~~Test config migration.~~

~~Test camera construction with one deliberately broken entry; expected result: bad camera is skipped, others continue.~~

~~Test missing model behavior and graceful degradation.~~

~~Test RKNN/ONNX metadata round-trip.~~

~~Power-cut test during config save.~~

Fresh boot -f test on Raspberry Pi and Orange Pi. (deferred: needs hardware)

~~Upgrade test from an old config.~~

~~End-to-end manual test: camera add/edit/delete, calibration wizard, model upload/select/delete, and multi-camera boot with one bad camera.~~

EXIT

CRITERIA

Tests actually fail when they should, pass when fixed, and the core release workflows survive the manual test matrix.

Day 6 - Multi-Board + Field Conditions

~~Give each board a unique mDNS hostname/suffix to prevent multi-coprocessor collisions.~~

~~Verify RKNN wheel/GitHub release URLs used by optimizer.py are live and version-correct.~~

Load-test with multiple MJPEG /video/ viewers and simultaneous camera streaming. (deferred: needs hardware)

Verify the SBC remains stable under realistic field load. (deferred: needs hardware)

~~Cross-platform smoke test on Windows, especially camera discovery.~~

~~Check RKNN target-platform detection: failed detection must be visible to the user instead of silently producing an rk3588-targeted artifact.~~

EXIT

CRITERIA

Two boards can coexist on the same network, core downloads work, streaming remains stable under load, and platform detection failures are obvious.

Day 7 - Docs + Packaging + Release Gate

~~Update README/install documentation to match the actual YoloModels directory structure.~~

~~Document known limitations and security/auth behavior.~~

~~Document final AGPL status and any architecture restrictions.~~

~~Update CHANGELOG/release notes with breaking config changes and upgrade requirements.~~

Run install.sh and install_windows.ps1 end-to-end on fresh environments. (scripts reviewed + syntax-checked; not run end-to-end)

Review all release-blocking and important audit findings one final time.

RELEASE

GATE

Only tag the release if all release blockers are closed or deliberately accepted with a documented, legally/technically valid release decision.

